

Characterizing Perfusion Assays in Neonatal Monitoring

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Abstract

Work in biomedical research has long relied on assumptions we test directly here. Residual error grows approximately linearly with the number of coupled parameters. We release the processing scripts and intermediate artefacts to support independent verification. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

1. Introduction

Replicate agreement was high, suggesting the protocol is robust to small handling differences. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Neither of the alternative formulations improved on the simple additive model in our setting. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Calibration drift over the trial period remained below the threshold set out in the protocol. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined. Residual error grows approximately linearly with the number of coupled parameters.

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2. Related Work

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3. Materials and Methods

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4. Experimental Setup

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5. Results

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses. Residual error grows approximately linearly with the number of coupled parameters. Sensitivity to the smoothing window proved modest across the range we examined. The measurements reported here were collected over twelve independent trials under controlled conditions. Neither of the alternative formulations improved on the simple additive model in our setting. Longer observation windows narrow the confidence interval but do not shift the central estimate.

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6. Discussion

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7. Limitations

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